

The Idea of a Base — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 · MCQ

[1]

In a base- n number system, the landmark numbers are:

(a) multiples of n (b) powers of n (c) factors of n (d) any numbers

Q2 · MCQ

[1]

The Egyptian number system is a:

(a) base-5 system (b) base-10 system (c) base-60 system (d) base-2 system

Q3 · FILL IN THE BLANK

[1]

The number 143 written in base 5 is _____.

Q4 · MCQ

[1]

The base-5 number 1033 equals which value in base 10?

(a) 143 (b) 153 (c) 128 (d) 1033

Q5 · TRUE / FALSE

[1]

In base 5, five of one landmark number make the next landmark number.

True False

Q6 · ASSERTION-REASON

[1]

Assertion (A): The number 100 is written as 400 in base 5.

Reason (R): $100 = 4 \times 25 = 4 \times 5^2$.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

[2]

Convert 324 (base 10) into base 5.

Q8

[2]

Convert the base-5 number 2123 into base 10.

Q9

[2]

Write the first six landmark numbers of a base-5 system.

Long answer (3 marks each)

Q10

[3]

Express the number 100 in (i) base 5 and (ii) base 4. Show your grouping.

Q11

[3]

In a base-4 system, write the number 16 and explain why it is written that way.

Q12

[3]

Explain, with an example, why a number system with a base makes multiplication easier than the Roman system.

Total: 21 marks · Check your work against the [answer key](#).